



INCT Signals

Signal Processing, Communications, Sensing & Surveillance

Airborne and Spaceborne Remote Sensing & Surveillance

A national research network connecting RF hardware, antennas, sensing platforms, signal processing, AI and autonomous systems.

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Supported by  **inct**
Institutos nacionais
de ciência e tecnologia

 **CNPq**
Conselho Nacional de Desenvolvimento
Científico e Tecnológico

 **CAPES**

 **FUNCAP**

National Institute of Science and Technology

INCT Signals at a glance

A multidisciplinary Brazilian network designed to turn advanced sensing and communications research into deployable technologies.

**5**

partner institutions

**6**

thematic axes

**4**

integrated work packages

**25**

core researchers

**94**published / accepted
papers

Leadership & coordination

Coordinator

André L. F. de Almeida · UFC

Vice-Coordinator

Felix D. Antreich · ITA

Local coordinators

César A. M. Marcon · PUCRS
Marcos V. T. Heckler · UNIPAMPA
Edison P. de Freitas · UFRGS

Mission

Research excellence • human-capital
development • technology creation •
national and international collaboration

From strategic challenges to practical impact

INCT Signals integrates sensing, communications and intelligent processing to create lower-cost, robust and nationally relevant solutions.



What we address

- Remote sensing and environmental monitoring
- Airborne and satellite surveillance
- Reliable wireless links and GNSS
- Integrated communications and sensing
- Autonomous multi-platform operation



How we work



Platforms & testbeds



RF, antennas & hardware



Signal/data processing



AI & autonomous systems



Why it matters



Strategic autonomy & sovereignty



Environmental & climate insight



Industrial innovation

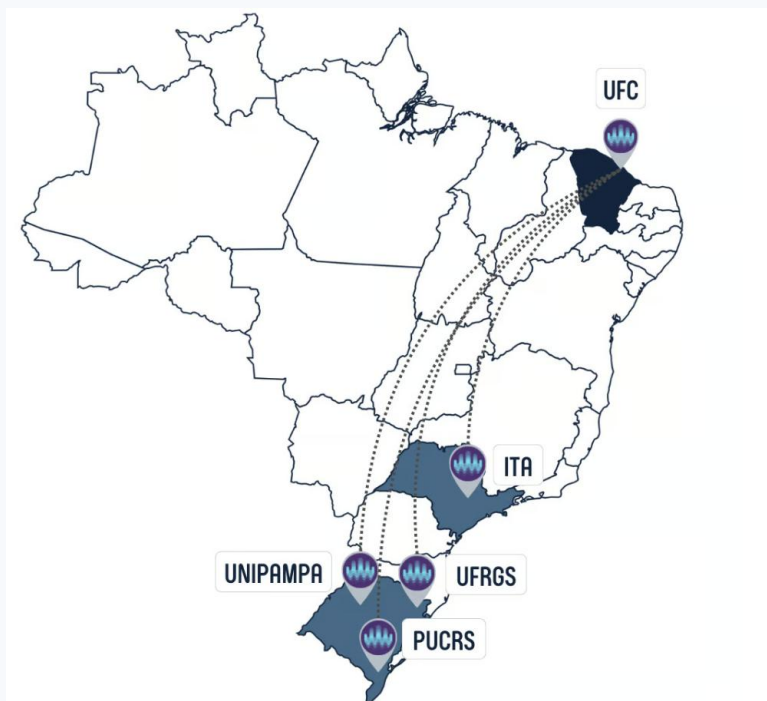


Highly qualified talent



Scientific & social impact

A national network of complementary capabilities



Partner institutions: **UFC** · **ITA** · **UFRGS** · **PUCRS** · **UNIPAMPA**

UFC

Signal processing, wireless communications, satellite microelectronics; algorithms, experiments and fault-tolerant satellite communications.

ITA

Remote sensing, GNSS, integrated circuits and satellite technologies; experimental platforms and sensing algorithms.

UFRGS

Multi-UAV control and navigation, satellite systems, fault tolerance and wireless integration.

PUCRS

Digital and embedded systems, manycore architectures, task mapping and fault-tolerant satellite processing.

UNIPAMP A

Antennas and microwave circuits for communications, satellite radionavigation and synthetic-aperture radar.

Six thematic axes connect the full sensing stack



**Communications,
signal processing &
optimization**



**Remote sensing &
radar systems**



**Environmental &
climate monitoring**



**Signal surveillance,
security & reliability**



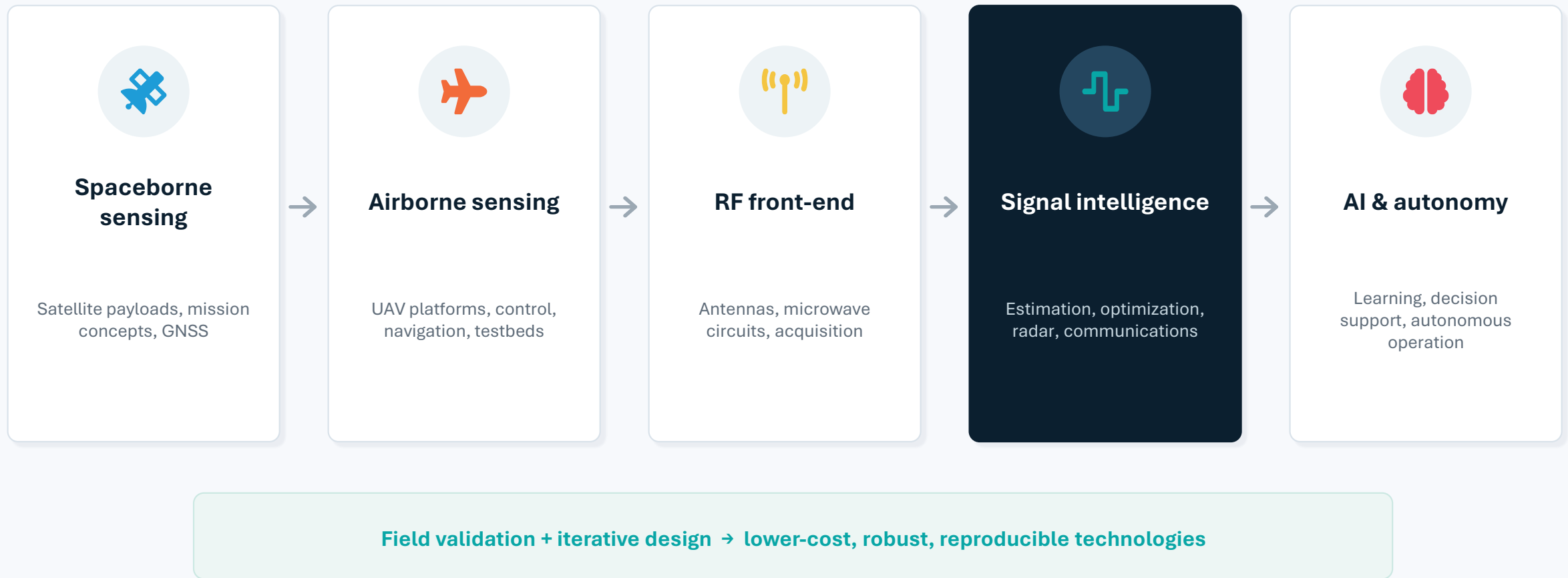
**Autonomous &
intelligent systems**



**Electromagnetic
characterization for
communications**

From physical platforms to actionable intelligence

The Institute is structured to close the loop between hardware, data acquisition, algorithms and field validation.



Four work packages organize the research program

WP1



Satellite-based remote sensing & surveillance

UFC • PUCRS

- WP1.1 Payload & components
- WP1.2 Mission concepts & planning

WP2



Hardware, antenna designs & RF development

UNIPAMPA • ITA • UFC

- WP2.1 Antennas
- WP2.2 RF front-end
- WP2.3 Signal recording & pre-processing

WP3



Testbeds & platforms for UAV-based sensing & surveillance

UFRGS • ITA • UFC

- WP3.1 UAV experimental platform
- WP3.2 UAV testbed

WP4



Signal processing for sensing & communications

UFC • ITA

- WP4.1 Joint communications & sensing
- WP4.2 Radar sensing signal processing

Core technology capabilities



RF & antennas

Microwave circuits, SAR antennas, reconfigurable patterns, electromagnetic characterization



Sensing platforms

UAV coordination, airborne sensing, satellite payloads, mission concepts and experimental testbeds



Signal processing

Tensor methods, channel/parameter estimation, radar processing, array processing and optimization



Communications & ISAC

Wireless links, GNSS, 5G/6G, joint communications and sensing, intelligent surfaces



Embedded & reliable systems

Manycore architectures, fault tolerance, satellite computing and signal acquisition



AI & autonomous systems

Federated learning, intelligent monitoring, autonomy and data-driven decision support

Training the next generation of researchers and engineers

Human-capital development is a core mission: students participate in multidisciplinary research, experiments, seminars, workshops, mobility and innovation activities.

30 undergraduate

6 MSc

10 PhD

4 postdoctoral
researchers

Development pathway



Undergraduate
research



Graduate research



Postdoctoral training



International mobility



Learning environment

- Hands-on testbeds and experimental platforms
- Annual project workshops and research seminars
- Scholarship opportunities in Brazil and abroad
- Interactions with international experts and IEEE lecturers

International research collaborations

The Institute combines a strong Brazilian core with research connections to universities and research centers abroad.



How external collaboration adds value

- **Joint scientific publications**
co-authorship across complementary expertise
- **Co-supervision & mobility**
students and researchers connected across institutions
- **Specialized expertise**
access to methods, know-how and complementary resources
- **Data & infrastructure**
experimental data, laboratories and validation environments

Formal cooperation agreements turn connections into joint action

Current agreements support joint research, mobility, access to specialized infrastructure, experimental validation and technology transfer.

**DLR****German Aerospace Center • Germany****Remote sensing • radar • navigation**

Airborne/spaceborne data • aerospace validation • geospatial signal processing

**CEDRA****Brazil****Digital agriculture • IoT • embedded systems**

Field validation • sensing for agriculture • connectivity • technology transfer

**UESTC****China****Wireless communications • sensing • radar**

Tensor modeling/estimation • signal processing • researcher training & internationalization

**Skoltech****Russia****Signal/data processing • AI • autonomous systems**

Machine learning • deep neural models • communications and space applications

Expected outcomes: joint projects • co-authored papers • mobility • shared infrastructure • prototypes • innovation

Connections with IEEE scientific societies

IEEE chapter links create a practical bridge between the project and global scientific communities in signal processing, communications, computing and photonics.



Local IEEE gateways



**IEEE Northeast Brazil
Signal Processing Society
Chapter**



**IEEE Northeast Brazil Joint
Chapter**

The Joint Chapter integrates ComSoc, Computer Society and Photonics Society.



Partner societies

SPS

IEEE Signal Processing Society

ComSoc

IEEE Communications Society

CS

IEEE Computer Society

IPS

IEEE Photonics Society



What these links enable

- Co-financing of events and outreach
- Distinguished Lecturers and invited experts
- Workshops, schools and seminar series
- Student and early-career researcher development
- New scientific collaborations and global visibility

Scientific output: a growing publication portfolio

Website snapshot · 7 August 2026. Published/accepted outputs are counted separately from records explicitly marked as submitted.



94

published / accepted
scientific papers

34 journal articles
+ 60 conference papers

Current website records



40

International
conference papers



34

Journal
articles



20

Brazilian / national
conference papers

Accepted/published conference split



Active pipeline

16 manuscript records
currently listed as
submitted

6 conference records
explicitly marked
submitted

Pipeline figures are not added to the 94
total and may change as status updates
are posted.

Visible scientific and technological momentum



14 papers presented at SBrT 2025

A strong national scientific presence across several project institutions



Best Paper Award in SBrT 2025

Undergraduate-led work on RIS-assisted channel estimation



3 PhD defenses highlighted in 2026

Rubem Pacelli, Amarilton Magalhães and Daniele Oliveira



IEEE Distinguished Lecture ecosystem

IEEE chapters and societies bring international experts into the project network

Dissemination channels include annual workshops, technical seminars, invited talks, project news, videos and publications.



Connect with INCT Signals

We welcome collaborations that connect methods, hardware, experimental platforms and real-world sensing challenges.



Joint R&D and proposals



Experimental validation & shared data



Student and researcher mobility



Seminars, workshops & technical exchange

EMAIL inctsignals@gmail.com **WEB** inct-signals.org

YOUTUBE [@INCTSignals](https://www.youtube.com/@INCTSignals)



WEB



VIDEO

Thank you

